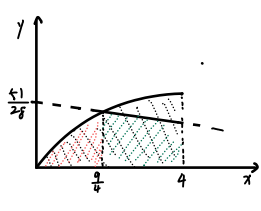


$$E(Y|X) = \int_{\mathbb{R}} f_{Y|X}(y|x) \cdot y \, dy$$


$$\begin{aligned} E(Y) &= \int_{\mathbb{R}} f_Y(y) \cdot y \, dy \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f_{Y|X}(y|x) \cdot f_X(x) \, dx \right) \cdot y \, dy \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f_{Y|X}(y|x) \cdot y \, dy \right) \cdot f_X(x) \, dx \\ &= E[E(Y|X)] \end{aligned}$$

$$Var(Y) = \int_{\mathbb{R}} f_Y(y) \cdot [y - E(y)]^2 \, dy$$

$$= \int_{\mathbb{R}} \int_{\mathbb{R}} f_{Y|X}(y|x) \cdot f_X(x) \, dx \cdot [y - E(y)]^2 \, dy$$

$$= \iint_{\mathbb{R}^2} f_{Y|X}(y|x) \cdot f_X(x) \cdot [y - E(y)]^2 \, ds = \int_{\mathbb{R}} f_X(x) \, dx \int_{\mathbb{R}} f_{Y|X}(y|x) \left([y - E(y)]^2 + [E(y) - E(Y)] \cdot [E(y|x) - E(Y)] \right) \, dy$$

$$E[Var(Y|X)] = \int_{\mathbb{R}} f_X(x) \cdot \int_{\mathbb{R}} f_{Y|X}(y|x) [y - E(y|x)]^2 \, dy \, dx$$

$$\begin{aligned} Var[E(Y|X)] &= \int_{\mathbb{R}} f_X(x) \cdot [E(Y|X) - E(E(Y|X))]^2 \, dx \\ &= \int_{\mathbb{R}} f_X(x) [E(Y|x) - E(Y)]^2 \, dx \end{aligned}$$

$$y_{n+1}(x) = y_0 + \int_{x_0}^x f(t, y_n(t)) \, dt.$$

$$y_{n+1}(x) - y_0(x) = \int_{x_0}^x [f(t, y_n(t)) - f(t, y_0(t))] \, dt$$

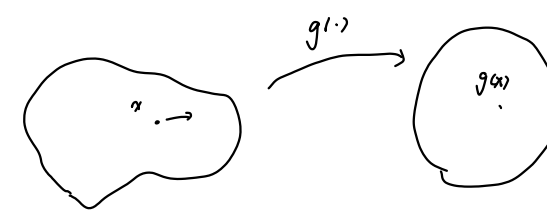
$$\|y_1 - y_0\| \leq M \|x - x_0\|$$

$$|x_n - \dots - x_{n+m}|$$

$$\text{suppose } E[\chi_i] = 0$$

$$\text{set } Y_i = \chi_i \cdot \mathbb{1}_{\{|x_i| \leq 1\}}$$

$$\begin{aligned} E[\chi_i] &= \int_{-\infty}^{\infty} x \, P(X=x) \, dx \\ &= \int_0^{\infty} x \, P(X=x) \, dx \\ &= \int_0^1 x \, P \, dx + \int_1^{\infty} x \, P \, dx + \dots \\ &\geq 0 \cdot \int_0^1 P \, dx + 1 \cdot \int_1^{\infty} P \, dx + \dots \\ &= \sum P(|x_i| > i) \end{aligned}$$



$$\lim_{\Delta x \rightarrow 0} \frac{e^{\Delta x} - 1}{\Delta x} = 1$$

$$[(e^x)' = e^x]_{x=0}$$

$$\frac{d}{dx} e^x = e^x$$